

Subject: Digital Signal and Image Processing (01CT1513)

Aim: cross correlation and auto correlation on discrete signal time signal.

Experiment No: 3

Date : 31-07-2026

Enrollment No:92400133030

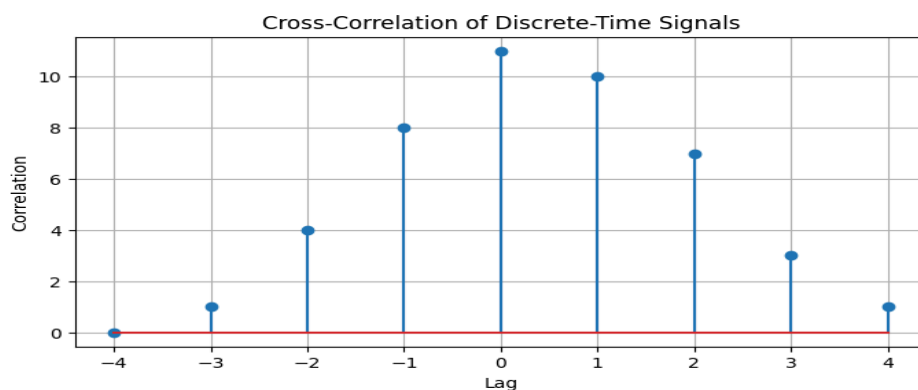
Aim: cross correlation and auto correlation on discrete signal time signal.

Code for CROSS CORRELATION on Discrete signal time signal :

```
import numpy as np
import matplotlib.pyplot as plt

# Discrete-time signals
x = np.array([1, 2, 3, 2, 1])
y = np.array([1, 1, 2, 1, 0])
# Cross-correlation
rxy = np.correlate(x, y, mode='full')
lags = np.arange(-(len(y)-1), len(x))
# Display output
print("Cross-Correlation:")
print(rxy)
# Plot only the cross-correlation
plt.figure(figsize=(8,4))
plt.stem(lags, rxy)
plt.title("Cross-Correlation of Discrete-Time Signals")
plt.xlabel("Lag")
plt.ylabel("Correlation")
plt.grid(True)
plt.show()
```

OUTPUT :



Code for AUTOCORRELATION on discrete signal time Signal.

```
import numpy as np
```

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```
import matplotlib.pyplot as plt

# Discrete-time signal
x = np.array([1, 2, 3, 2, 1])

# Autocorrelation
auto_corr = np.correlate(x, x, mode='full')
lags = np.arange(-(len(x)-1), len(x))

# Display output
print("Autocorrelation:")
print(auto_corr)

# Plot only the autocorrelation
plt.figure(figsize=(8,4))
plt.stem(lags, auto_corr)
plt.title("Autocorrelation of Discrete-Time Signal")
plt.xlabel("Lag")
plt.ylabel("Correlation")
plt.grid(True)
plt.show()
```

OUTPUT :

